Rosen — §6.2: The Pigeonhole Principle

Selected Exercises

Alston

Semester 2, 2026

Exercise 1

Show that in any set of six classes, each meeting regularly once a week on a particular day of the week, there must be two that meet on the same day, assuming that no classes are held on weekends.

Exercise 5

Show that among any group of five (not necessarily consecutive) integers, there are two with the same remainder when divided by 4.

Exercise 7

Let n be a positive integer. Show that in any set of n consecutive integers there is exactly one divisible by n.

Exercise 8

Show that if f is a function from S to T, where S and T are finite sets with $|S| > |T|$, then there are elements s_1 and s_2 in S such that $f(s_1) = f(s_2)$, or in other words, f is not one-to-one.

Exercise 9

What is the minimum number of students, each of whom comes from one of the 50 states, who must be enrolled in a university to guarantee that there are at least 100 who come from the same state?

Exercise 11

Let (x_i, y_i, z_i), $i = 1, 2, 3, 4, 5, 6, 7, 8, 9$, be a set of nine distinct points with integer coordinates in xyz space. Show that the midpoint of at least one pair of these points has integer coordinates.

Exercise 21

Construct a sequence of 16 positive integers that has no increasing or decreasing subsequence of five terms.

Exercise 22

Show that if there are 101 people of different heights standing in a line, it is possible to find 11 people in the order they are standing in the line with heights that are either increasing or decreasing.

Exercise 24

Suppose that 21 girls and 21 boys enter a mathematics competition. Furthermore, suppose that each entrant solves at most six questions, and for every boy-girl pair, there is at least one question that they both solved. Show that there is a question that was solved by at least three girls and at least three boys.

Exercise 29

Recall: Ramsey numbers were discussed after Example 13 in Section 6.2. $R(m, n)$ is the smallest integer such that any two-coloring of the edges of the complete graph $K_{R(m,n)}$ contains either a red K_m or a blue K_n.

Show that if n is an integer with $n \geq 2$, then the Ramsey number $R(2, n)$ equals n.

Exercise 40

Prove that at a party where there are at least two people, there are two people who know the same number of other people there.

Exercise 41

An arm wrestler is the champion for a period of 75 hours. (Here, by an hour, we mean a period starting from an exact hour, such as 1 p.m., until the next hour.) The arm wrestler had at least one match an hour, but no more than 125 total matches. Show that there is a period of consecutive hours during which the arm wrestler had exactly 24 matches.

Exercise 43

Show that if f is a function from S to T, where S and T are nonempty finite sets and $m = \lceil |S|/|T| \rceil$, then there are at least m elements of S mapped to the same value of T. That is, show that there are distinct elements $s_1, s_2, \ldots, s_m$ of S such that $f(s_1) = f(s_2) = \cdots = f(s_m)$.